

| Questions from Interested Parties |                                                                                                                                                                                               | Answers from NASA                                                                                                                                                                                                                                                                                                                                                                                             |
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| 1                                 | What are the outcomes that NASA desires from acceleration of flight testing under the AFA program?                                                                                            | NASA is looking to utilize it's subject matter expertise and technical capability assets to make flight testing more accessible as a way of concurrently gathering data, identifying and reducing risks that may not be apparent from modeling and ground test work, and rapidly maturing technology and integrated aircraft concepts that advance the nation.                                                |
| 2                                 | What resources does NASA plan to have available to support these activities?                                                                                                                  | NASA's subject matter experts, test facilities including the AETC and center assets, aircraft and test ranges, and various laboratories across the four aero centers (Ames, Armstrong, Glenn, and Langley) are available to support these activities. A more complete listing of these are available in the industry day slide packages. NASA may also contribute capital as part of a cost sharing activity. |
| 3                                 | What does success look like from a NASA perspective?                                                                                                                                          | NASA aims to accelerate technology development and learnings from flight research and testing that will enable evolutionary advancements in support of the US Aeronautics industry. Ideally the AFA would support a diverse array of activities that included partnerships with large and small partners across many aeronautics technology areas.                                                            |
| 4                                 | Does NASA share the view that potential content complementary to solicitation #80GRC0026R002 is beneficial to accelerating open fan propulsion for future commercial products?                | NASA's Aeronautics Flight Accelerator will not show preference for any perspective partner based on past or current agreements.                                                                                                                                                                                                                                                                               |
| 5                                 | Which NASA-operated or partnered large drone platforms have the payload capacity and power infrastructure suitable for testing a 2-2.5 MW class powertrain subsystem (motor, drive, thermal)? | Additional details of the 2-2.5 MW class powertrain subsystem (motor, drive, thermal) will need to be provided to determine the best NASA or other government agency platform.                                                                                                                                                                                                                                |

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| 6 | What are the established processes and timelines for obtaining airworthiness approval to integrate and fly experimental, high-power electrical systems on NASA drone testbeds?                                                   | <p>Researchers proposing work on NASA testbeds or a “drone testbed” should start the conversation with the owner of that testbed, because they likely have the best insight of their airworthiness approval process. If the testbed owner doesn't have this insight, they are encouraged to engage the appropriate airworthiness authority early, in order to bake in airworthiness concerns into the development and/or integration plan. At Armstrong, the airworthiness authority resides with the AFRC Chief Engineer’s Office. It's also important to note that, while flying on a UAS can allow a project to buy a lot of risk, this approach comes with challenges that aren't necessarily mitigated by an uninhabited aircraft. If the experimental high-power electrical system has a high likelihood of destroying the testbed, or causing it to destroy itself, then some kind of agreement needs to be brokered between the project and the testbed owner, to inform the airworthiness approach. Regardless of project, the airworthiness story should be developed during the engineering development phase; that airworthiness approach will then be followed during the system integration onto the aircraft.</p> <p>The NASA Airworthiness Process is not a checklist like an FAA certification would be. It is a risk-based process, which means that it operates on the idea that the project and any independent reviewers are able to assess the risk and present it in an Airworthiness and Flight Safety Review process. If the level of risk of the realized system is at the same level as the project is able to accept, or if the project has negotiated a level of acceptance at a higher approval level (in AFRC's typical process, that would be the AFRC Center Director), then they can go to flight. Two examples to illustrate level of rigor and independent review: a) If the system simply operates in flight, doing its own thing with very little interaction with the airplane and poses only a small amount of additional risk to an operation, the airworthiness process can be as simple as a Tabletop review held in the Chief Engineers' office, leading to a flight release; b) If the system represents a major modification to the airframe, affecting multiple critical systems and substantially increases risk to personnel or high-value assets, then the project is likely to have to go over their flight test goals and plans with an independent review board (known as a Flight Readiness Review (FRR) at AFRC), who will then present to the Airworthiness and Flight Safety Review Board (known as the AFSRB) in order to obtain authority to proceed to Flight Phase. There is a continuum between those two extremes regarding rigor and independent review.</p> |
| 7 | Can the AFA program facilitate early-stage discussions with FAA representatives to address key certification questions for MW-scale electric propulsion systems for larger aircraft, based on the data gathered in this program? | <p>NASA has a long history of collaboration with the FAA on many activities and to overcome many challenges. Key learnings from many of the aeronautics projects and programs have informed FAA rule making and practices. The results from work in the AFA, provided data rights allow, would be available to the FAA and could be used in the described early-stage discussions. Facilitating these discussions also aligns with NASA ARMD's mission.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

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| 8 | What specific flight test instrumentation and data acquisition capabilities are available on NASA's drone platforms to support detailed characterization of a novel electric powertrain? | <p>Regarding data acquisition and instrumentation, it is absolutely possible to instrument UAS. For small UAS, these options range from smaller, self-contained COTS packages that can be purchased and mounted (per the available mounting locations and weight constraints), to even smaller/lighter packages that are custom-made to fit within weight/data/testbed requirements. For a larger integration, it comes down to limiting the effect of the added hardware on the UAS platform itself. The scale and complexity of these integrations really depend on the testbed and the experiment. Without those details, the answer is quite often "it depends".</p> <p>Another aspect for projects and researchers to consider is how to build mitigations (to protect the testbed) into the experiment itself. For example, can the experiment be quickly turned off (via quick-burn fuses)? Could a firebox be built around the experiment? Can the equipment be jettisoned? Thinking early about these kinds of mitigations and ways to protect the aircraft/testbed from the experiment can lower the risk of the experiment and make the flight experiment easier if those mitigations are considered early in the experiment design. Bottom line - projects will benefit from adopting a philosophy of "how can we protect the aircraft/testbed" from the very beginning of experiment development.</p> |
| 9 | Available flight platforms and test assets?                                                                                                                                              | <p>The list of flight platforms and test assets are available in the Industry Day presentations from AETC, and the four research centers.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

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| 10 | Airworthiness requirements for participation? | <p>The NASA Airworthiness Process is not a checklist like an FAA certification would be. It is a risk-based process, which means that it operates on the idea that the project and any independent reviewers are able to assess the risk and present it in an Airworthiness and Flight Safety Review process. Regardless of project, the airworthiness story should be developed during the engineering development phase; that airworthiness approach will then be followed during the system integration onto the aircraft. If the level of risk of the realized system is at the same level as the project is able to accept, or if the project has negotiated a level of acceptance at a higher approval level (in AFRC's typical process, that would be the AFRC Center Director), then they can go to flight. Two examples to illustrate level of rigor and independent review: a) If the system simply operates in flight, doing its own thing with very little interaction with the airplane and poses only a small amount of additional risk to an operation, the airworthiness process can be as simple as a Tabletop review held in the Chief Engineers' office, leading to a flight release; b) If the system represents a major modification to the airframe, affecting multiple critical systems and substantially increases risk to personnel or high-value assets, then the project is likely to have to go over their flight test goals and plans with an independent review board (known as a Flight Readiness Review at AFRC), who will then present to the Airworthiness and Flight Safety Review Board (known as the AFSRB) in order to obtain authority to proceed to Flight Phase. There is a continuum between those two extremes regarding rigor and independent review. The project is strongly encouraged to engage the appropriate airworthiness authority early, in order to bake in airworthiness concerns into the development and/or integration plan. At Armstrong, the airworthiness authority resides with the AFRC Chief Engineer's Office.</p> |
| 11 | Hybrid-electric architecture priority?        | <p>NASA is exploring next generation propulsion systems including hybrid propulsion technologies to enable new types of aircraft. Disciplines are: 1) Hybrid Propulsion methods to advance operability, performance, and safety for complex propulsion systems 2) Energy Storage Advancing alternative energy storage research for electric aviation. Specific investments are being made in integrated flight, power, and engine control for distributed electric propulsion, energy storage, thermal systems, and electric machines and drives. This activity is included among a suite of solutions that NASA considers crucial to the future aviation industry.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 12 | Specific Pain points desired to be addressed? | <p>NASA is seeking revolutionary concepts and solutions across all flight regimes. This RFI is not focused on any specific pain points, but is focused on providing opportunities to utilize NASA subject matter expertise, ground test capabilities, and flight research capabilities and assets to aid in technology research and development for the benefit of the United States.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 13 | Preferred/ required cost-share ranges?        | <p>Any cost share preferences will be identified in future requests for proposals.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

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| 14 | What data rights are expected for this contract?                                                                                                             | At this time, NASA has not yet determined the acquisition/funding approach (e.g., SBIR/STTR, FAR-based contract, or a funded Space Act Agreement under the National Aeronautics and Space Act), so specific data rights and intellectual property terms are not yet defined. In general, entities that partner with NASA typically retain ownership of pre-existing intellectual property, and rights in data and inventions developed under the effort follow the framework associated with the selected instrument. NASA typically seeks a Government-purpose license to use data generated under the effort to support agency missions, including retaining data for archival purposes and limited Government use. As is customary, NASA may also include provisions that allow broader Government use under certain conditions, such as when necessary to ensure practical application, continuity of benefits, or to address mission, public use, or safety needs. Additional details will be provided if/when the acquisition strategy is finalized. |
| 15 | Please confirm whether a Cover page, Table of Contents, or other information is requested in the response? If so, will these count against the 8 page limit? | A cover page and table of contents are not required; however, if included, they will not be counted toward the 8-page limit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |